

MECHANICAL BEHAVIOR AND STRUCTURAL PERFORMANCE OF A Sn-CuSn-Pb COATED AND EPOXY COMPOSITE PLAIN BEARING

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ABSTRACT

Plain bearings, such as those used in crankshafts and connecting rod bearings in the engine block, are an essential component of many mechanical systems. They are characterized by a simple design, comprising a fixed bearing surface and a journal, which is the part of the shaft in direct contact with the bearing, ensuring sliding on the bearing surface. To minimize friction and wear, a continuous supply of lubricating oil is required. However, if the oil pressure is insufficient or if the quality of the oil used does not meet the specific requirements in terms of viscosity and rheological properties, plain bearings can wear prematurely, leading to potentially serious engine failures.

To remedy these problems and improve the durability of plain bearings, technological advances have been made. Engineers have developed coated plain bearings and even composite material bearings. These innovative designs aim to reduce wear, increase the reliability of internal combustion engines, and ensure that the whole engine is lightweight.

This numerical study aims to carry out an in-depth analysis of the structural behaviour of two types of plain bearing used in internal combustion engines: coated-based tin (Sn,) bronze-based (Cu), and lead-based (Pb), and those made of epoxy composite.

The main objective is to provide accurate predictions of the elastic behaviour of these structures under different operating conditions. The structural parameters examined include displacement and shear stress concentration at the inner

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surface of the bearings. To achieve this objective, a numerical approach is adopted, using the finite element method to solve the displacement equations. This method allows detailed modeling of the mechanical behaviour of plain bearings, taking into account the material properties of the Sn-Cu-Pb coatings and epoxy composites, as well as the load and rotational speed conditions.

The results obtained highlight several important conclusions. It was observed that the shear stresses were more pronounced for the bronze-coated bearings, particularly at a rotational speed of 6,000 rpm. The elastic deformations are higher for epoxy composite plain bearings, compared with those observed for a coated bearing.

Keywords: plain bearing, coated, composite, mechanical behaviour, internal combustion engine.

AIMS AND BACKGROUND

Cha et al.¹ showed the influence of bearing support elasticity, bearing support geometry and bearing liner properties on the dynamic characteristics of the bearing. Despite a lot of scientific research showing the effect of the design parameters of the swing pad bearings, i.e., the position of the pivot, the type of pivot and the shape of the swing pad on the dynamic properties, there is little information available on the effect of the materials used for the pad coating on the performance characteristics of the bearings. White metal has been widely used as a bearing lining material due to its low cost, good conformability, and integration capability.

However, in the case of insufficient oil film formation or unexpected oil breakage between the bearing and the rotor, conventional white metal bearings fail immediately, causing serious damage to the rotor². It is known that metallic materials have poor damping properties, and therefore white metals can fail under the fatigue load induced by the rotating shaft. Fatigue stress plays an important role in the failure of white metal oscillating pads due to vibration. To improve the damping properties, Simmons et al.³ introduced polymer-faced tilting pads (PEEK and PTFE).

Maria et al.⁴ studied the influence of fibre length on the tribological properties of self-lubricating polymer composites. The author composed of PTFE (matrix), glass fibres and MoS₂. This study was carried out using the discrete element method (DEM). They found that a combined experimental/digital approach with DEM.

Maksimkin et al.⁵ studied the tribological performance of bulk-oriented UHMWPE/FMWCNT nanocomposites and metal-polymer composite sliding bearings under dry friction conditions at different loads and sliding speeds. They showed that the wear intensity for the metal-polymer composite sliding bearing

without FMWCNT was more than twice that of the metal-polymer composite sliding bearing with 1% FMWCNT.

Seung Yoon et al.⁶ presented a modal analysis to investigate the dynamic stability of plain tilting pad bearings made of carbon fibre/epoxy and reinforced metal composites. These bearings were manufactured to improve dynamic performance and durability. They showed that the composite bearings effectively reduce rotor vibration and increase bearing stability and durability compared to the white metal tilting pad bearing. They indicate that the dynamic cross-coupling coefficients of the white metal bearing were significantly lower than those of the composite bearing.

Sri Endah Susilowati et al.⁷ examined the effects of graphite weight on the mechanical properties of a Cu-Sn-Zn/Gr composite and found that wear rate and porosity values decreased but hardness, density, and yield strength values increased in the 0.5 and 1.0 weight percentages of graphite. A Cu-Sn-Zn-based composite with added graphite was developed as a self-lubricating material for plain bearings using powder metallurgy and sintering.

Bronze bearings with graphite additions are suitable for heavy loads, high loads, and high temperatures, lubrication is provided by adding graphite during the manufacturing process⁸. The material of the bronze bearings consists of 90% Cu-10% Sn with an additional 1.5% graphite. Addition of graphite to reduce friction during application^{9,10}. Furthermore, the addition of graphite to copper-tin alloys has shown that graphite-containing composites produced by introducing PM and TiC between the graphite and bronze can effectively improve the mechanical and tribological properties of the composites¹¹.

In 2020, Mehala et Bendaoud¹² demonstrated the elastic behaviour of a plain bearing coated with a textured surface and a non-textured surface. Surface texturing technology has recently been explored in the field of tribology and this method has been used to improve the displacement and performance of a Babbitt-metal journal bearing with a textured surface. A numerical analysis was carried out to study the effect of the textured surface on the elastic behaviour of the bearing. First, the bearing was tested without texture. Its behaviour was then examined after the application of a texture. Its performance was studied to develop a pressure, displacement and shear stress profile, generated for each combination of radial load and shaft speed. Numerical modeling of the bearing displacement under severe operating conditions was undertaken by solving the displacement equations using the finite element method. The results show that the elastic deformations of the textured bearings are large and predominate at high rotational speeds. Shear stress was also found to be significant at higher hydrodynamic pressures.

Hydrodynamic bearings are components of rotating machines subject to increasingly severe operating conditions. Their main purpose is to support radial

loads. Maria et al.¹³ studied the tribological performance of a homogeneous thermoplastic and thermoset composite in contact with stainless steel. They found that thermoset composites exhibit a higher level of static and dynamic friction compared to thermoplastic composites.

Alexey Rodichev et al.¹⁴ presented an experimental study of the wear resistance of two types of coatings (BrOF 10-1 and a molybdenum-based coating) and found that the molybdenum-based coating had a lower coefficient of friction and higher wear resistance than the bronze-based coating.

The essential objective of this analysis is to provide an in-depth understanding of the mechanical behavior of plain bearing structures, whether lined or composite. The study aims to accurately predict several significant aspects, including displacement, elastic deformation and shear stress concentration. Focusing on hydrodynamic plain bearings, this analysis explores how these mechanical parameters change with variations in operating conditions, such as rotational speed and radial load. These conditions are essential because they directly influence the stresses and forces applied to the bearing during operation of an internal combustion engine.

To reach this objective, the finite element method is used, offering a robust and accurate approach to model the behavior of structures under various conditions. This method allows efficient numerical resolution of the displacement equations, enabling bearing deformations, stresses and displacements to be calculated with a high degree of accuracy. The information obtained will be essential for engineers and designers of internal combustion engines, enabling them to optimize the design of plain bearings to ensure optimum performance, increased durability and improved reliability under a variety of operating conditions.

METHOD AND THEORY

Displacement Equations in 3D. In the theory of elasticity, solids are said to be linear elastic, meaning that once the stress that produced the displacement is restrained, the solid returns to its original shape. The classical equation of elasticity is used for displacement calculations, considering that solids are isotropic: all properties of a point are the same in all directions.

Suppose the particle $P(x,y,z)$ of element (e) – such as the tetrahedron shown in Fig. 1 – has a displacement of 0 initial state and 1 final state. The three (3) components of the displacement vector are approximated by 3 interpolation or displacement functions¹⁵.

Each node propagates 3 displacement components, so tetrahedral elements propagate 12 degrees of freedom (24 ddl propagation for 8-node hexahedral elements, the elements are hexahedral elements with 3 displacements per node).

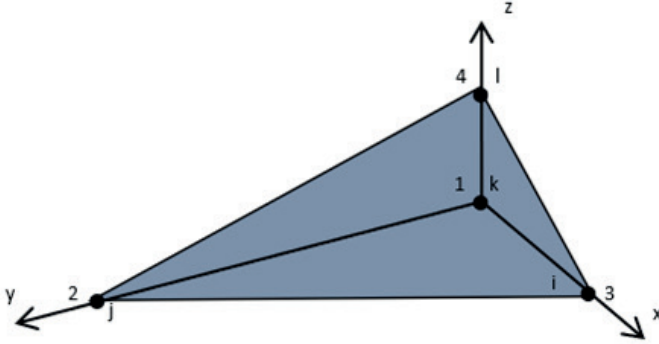


Fig. 1. Representation of a tetrahedral element¹⁵

The simplest functions of the displacement are of the order 1.

$$\overrightarrow{P_0 P_1} \begin{cases} u(x, y, z) = a_0 + a_1 x + a_2 y + a_3 z \\ v(x, y, z) = b_0 + b_1 x + b_2 y + b_3 z \\ w(x, y, z) = c_0 + c_1 x + c_2 y + c_3 z \end{cases} \quad (1)$$

By successively locating at the nodes i, j, k, l , the coefficients a_0, a_1, c_3 are expressed as a function of the 12 displacements of the nodes $u_i, v_i, \dots, w_l$.

$$\begin{cases} u_i = a_0 + a_1 x_i + a_2 y_i + a_3 z_i \\ \vdots \\ u_l = a_0 + a_1 x_l + a_2 y_l + a_3 z_l \end{cases} \quad \begin{cases} v_i = b_0 + b_1 x_i + b_2 y_i + b_3 z_i \\ \vdots \\ v_l = b_0 + b_1 x_l + b_2 y_l + b_3 z_l \end{cases} \quad \begin{cases} w_i = c_0 + c_1 x_i + c_2 y_i + c_3 z_i \\ \vdots \\ w_l = c_0 + c_1 x_l + c_2 y_l + c_3 z_l \end{cases} \quad (2)$$

By replacing the values of the coefficients, the displacement vector can be written as:

$$\overrightarrow{P_0 P_1} \begin{cases} u(x, y, z) = \alpha_i u_i + \alpha_j u_j + \alpha_k u_k + \alpha_l u_l \\ v(x, y, z) = \alpha_i v_i + \alpha_j v_j + \alpha_k v_k + \alpha_l v_l \\ w(x, y, z) = \alpha_i w_i + \alpha_j w_j + \alpha_k w_k + \alpha_l w_l \end{cases} \quad (3)$$

with $\alpha_i, \alpha_j, \alpha_l$ are the weighting function for the element (e).

Matricially, the displacement of a point amounts to:

$$\begin{Bmatrix} u \\ v \\ w \end{Bmatrix} = \begin{pmatrix} \alpha_i & 0 & 0 & \alpha_j & 0 & 0 & \alpha_k & 0 & 0 & \alpha_l & 0 & 0 \\ 0 & \alpha_i & 0 & 0 & \alpha_j & 0 & 0 & \alpha_k & 0 & 0 & \alpha_l & 0 \\ 0 & 0 & \alpha_i & 0 & 0 & \alpha_j & 0 & 0 & \alpha_k & 0 & 0 & \alpha_l \end{pmatrix} \begin{Bmatrix} u_i \\ v_i \\ w_i \\ \vdots \\ w_l \end{Bmatrix} \quad (4)$$

$$\begin{Bmatrix} u \\ v \\ w \end{Bmatrix} = \{\phi\}\{q\} \quad (5)$$

With $\{\phi\}$ Interpolation matrix

$\{q\}$ Displacement vector

The nodal forces equivalent to the surface force (pressure) is given by:

$$\{F_s\} = \iint_{S_e} [\phi]^t \{\vec{F}_s\} ds \quad (6)$$

The general fundamental equation that links nodal displacements $\{q\}$ and nodal forces is given by the following relationship:

$$\{F\} = \{K\}\{q\} \quad (7)$$

with $\{K\}$ is rigidity matrix.

Classical Laminate Theory. The theory of laminates concerns only thin structures by stratification of successive layers which results from the design of composite parts. In general, a plate is a solid limited by two parallel planes, with a small transverse dimension compared to the other two dimensions. It is then possible to define a reference plane between the two extreme planes which is taken as the Oxy plane. The Oz axis corresponds to the z direction perpendicular to the xy plane, according to the thickness (Fig. 2).

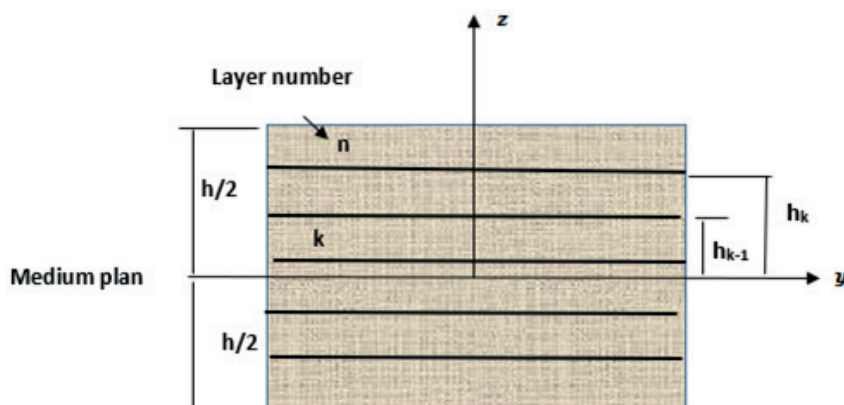


Fig. 2. Laminated plate

Assumptions of the classical laminate theory. The classical laminate theory uses a first-degree deformation scheme. It then makes an additional assumption that transverse shear is neglected. In this scheme, the transverse shear deformations are therefore zero, i.e.¹⁶:

$$\gamma_{xz} = \gamma_{yz} = 0$$

This assumption implies.

$$\varphi_x = -\frac{\partial w_0}{\partial x} \quad \varphi_y = -\frac{\partial w_0}{\partial y} \quad (8)$$

The displacement field can be written as

$$\begin{cases} u(x, y, z) = u_0(x, y) + z\varphi_x(x, y) \\ v(x, y, z) = v_0(x, y) + z\varphi_y(x, y) \\ w(x, y, z) = w_0(x, y) \end{cases} \quad (9)$$

The deformation field is written and taking into account the displacement expressions.

$$\begin{cases} \varepsilon_{xx} = \frac{\partial u}{\partial x} = \frac{\partial u_0}{\partial x} + z \frac{\partial \varphi_x}{\partial x} \\ \varepsilon_{yy} = \frac{\partial v}{\partial y} = \frac{\partial v_0}{\partial y} + z \frac{\partial \varphi_y}{\partial y} \\ \varepsilon_{zz} = \frac{\partial w}{\partial z} = \frac{\partial w_0}{\partial z} = 0 \end{cases} \quad (10)$$

$$\begin{cases} \gamma_{xy} = 2\varepsilon_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \\ \gamma_{xz} = 2\varepsilon_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \\ \gamma_{yz} = 2\varepsilon_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \end{cases} \Rightarrow \begin{cases} \gamma_{xy} = \left(\frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} \right) + z \left(\frac{\partial \varphi_x}{\partial y} + \frac{\partial \varphi_y}{\partial x} \right) \\ \gamma_{xz} = \varphi_x(x, y) + \frac{\partial w_0}{\partial x} = 0 \\ \gamma_{yz} = \varphi_y(x, y) + \frac{\partial w_0}{\partial y} = 0 \end{cases} \quad (11)$$

Shear results. The shear results are defined in the same way by:

$$Q(x, y) = \int_{-h/2}^{h/2} \sigma_k(M) dz \Rightarrow \begin{cases} Q_x = \int_{-h/2}^{h/2} \sigma_{xz} dz \\ Q_y = \int_{-h/2}^{h/2} \sigma_{yz} dz \end{cases} \Rightarrow Q(x, y) = \{Q_x\} = \int_{-h/2}^{h/2} \{\sigma_{xz}\} dz \quad (12)$$

Like the membrane results, the shear results are defined per unit length of the laminate. The discontinuity of stresses from one layer to the next leads to the above relationship being rewritten in the following form:

$$\begin{Bmatrix} Q_x \\ Q_y \end{Bmatrix} = \sum_{k=1}^n \int_{h_{k-1}}^{h_k} \begin{Bmatrix} \sigma_{xz} \\ \sigma_{yz} \end{Bmatrix}_k dz \quad (13)$$

NUMERICAL ANALYSES

The plain bearing was 100 mm in diameter and 80 mm in length. Its main characteristics are shown in Table 1. The bearing has two layers: the largest layer is stainless steel and 38 mm thick, and the inner layer is a coating with a thickness of 2 mm (see Fig. 3a). This pad was fed through a 70 mm long feed groove using three 14 mm diameter feed holes. Figure 3b shows a composite sliding bearing which is a laminate of two layers one mm thick with orientations $[0^\circ, 45^\circ]$.

Plain bearings are designed in coating and even in composite materials, in order to reduce wear, ie to increase the reliability of internal combustion engines, as well as to ensure the lightness of the engine. The purpose of applying a coating is to improve the surface properties of a mechanical part. For example, coatings can be used to preserve or enhance the appearance, adhesion, corrosion resis-

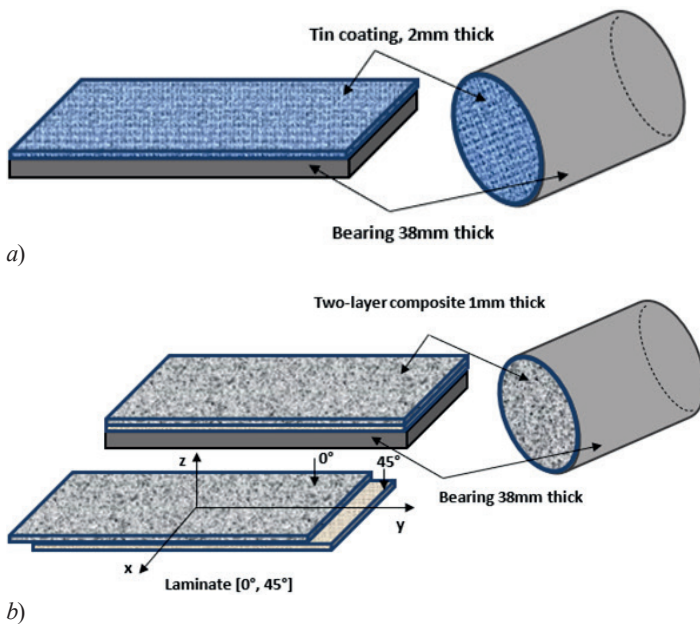


Fig. 3. Geometric representation of the plain bearing: a) Plain bearing coated with 88% tin, b) Composite plain bearing

Table 1. Characteristics of a plain bearing

	Tin Sn	Lead Pb	Bronze CuSn
Young's modulus, GPa	58.2	45.8	137.8
Tensile modulus, GPa	49.9	16.5	129.8
Density, kg/m ³	7280	11370	8960
Poisson coefficient	0.357	0.44	0.343

Table 2. Mechanical characteristics of different coatings

Elements	Dimension
Shaft diameter	99.82 mm
Bearing diameter	100 mm
Ring diameter	140 mm
Bearing length	80 mm
Radial clearance	90 mm
Bearing thickness	20 mm
Slot length	70 mm
Groove width	16 mm
Feed port diameter	14 mm

Table 3. Mechanical characteristics of different composites

Property	Boron / epoxy	Graphite / epoxy	Epoxy carbon UD
E_{1^*} , GPa	200	150	121
E_{2^*} , GPa	19.6	9	8.6
E_{3^*} , GPa	19.6	9	8.6
ν_{12}	0.3	0.3	0.27
ν_{13}	0.28	0.3	0.4
ν_{14}	0.28	0.01	0.27
G_{12^*} , GPa	7.2	7.1	4.7
G_{13^*} , GPa	5.5	7.1	3.1
G_{14^*} , GPa	5.5	2.5	4.7
Density	2000	1580	1490

tance, wettability, or adjust the properties of the surface of a part in the face of mechanical constraints and the external environment (ultraviolet rays, water, oxidation (corrosion), temperature, and mould, etc.). Table 2 shows the mechanical characteristics of the different coatings, and Table 2 shows the mechanical characteristics of the different composites.

In this part, the study focuses on the impact of hydrodynamic pressure on the electrohydrodynamic behaviour of a bearing subjected to high loads. We also examine the influence of parameters such as pressure, load and rotational speed on the displacement of the system using CFD code.

The model design software used can operate like an existing computer-aided model (CAD). This parametric volumetric modeling is supported by advanced technological features that allow quick and easy sketching, 3D modeling of parts, or downloading of CAD models for pre-processing using the finite element method. Figure 4 shows the geometry of an untextured plain bearing, providing a visual overview of the structure under study.

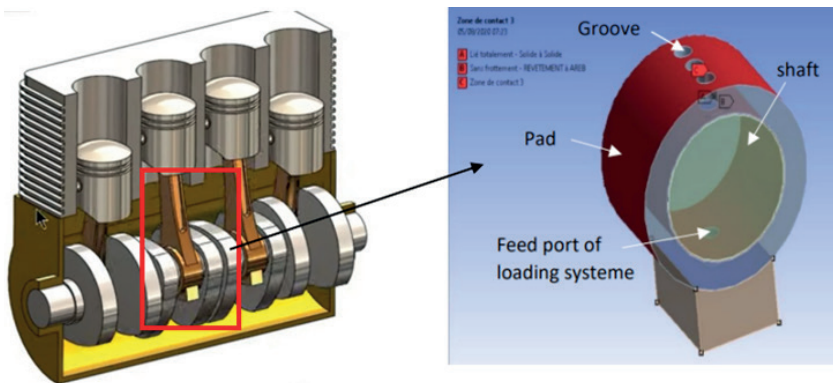


Fig. 4. 3D structure of the plain bearing.

The mesh. The pad is decomposed into 4 nodes using tetrahedral elements, this choice being dictated by the geometric complexity of the groove comprising lines, arcs, circular holes and spherical textures. These geometric features require the use of tetrahedral elements to ensure precise mesh overlap (see Fig. 5). The bearing is discretized into 12 points in the axial direction, while its angular amplitude is defined by 70 points and 4 nodes, adjusted for thickness. The feed groove is discretized into 13 points in the radial direction, with 5 nodes covering its width.

The overall mesh is made up of 19.830 elements and 47.784 nodes for a flat surface. This grid size provides an accurate representation of the complex geometry of the system under study, ensuring reliable and accurate numerical simulation of the physical phenomena involved.

In the present study, a range of structural and thermal loads can be applied, as well as imposed displacements. In the specific case examined here, uniform pressures were applied along the circumferential axis of the shaft and bearing (see Fig. 6). This allows the forces and stresses that occur under real operating

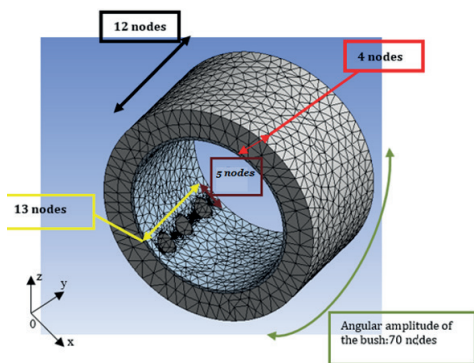


Fig. 5. Mesh of the plain bearing

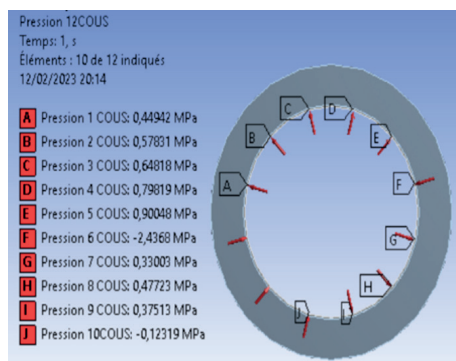


Fig. 6. Representation of pressures

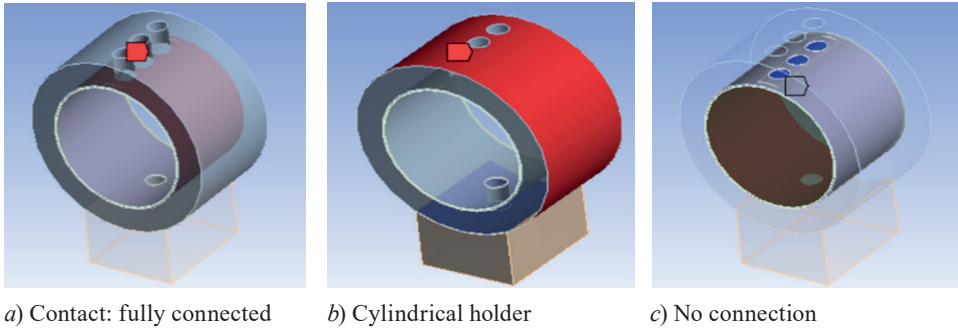


Fig. 7. Boundary conditions.

conditions to be simulated, providing an accurate representation of the structural behavior of the system.

The 2mm thick coating was integrally bonded to a 38 mm pad of base material (see Fig. 7a). This process ensures a strong and uniform bond between the coating and the substrate, ensuring optimum performance of the system as a whole.

The coated pad was positioned on a cylindrical contact support (see Fig. 7b), providing a secure and stable installation for the current experiment or analysis. This configuration also ensures that loads and stresses are evenly distributed over the entire surface of the pad.

Figure 7c highlights the separation of the rotating shaft from the liner. This separation is important to allow rotary movement of the shaft while maintaining the integrity of the liner and minimizing friction and excessive stresses that could compromise the operation of the system.

Model validation. In this section, a comparison is made between the numerical results obtained and those obtained experimentally, as documented in the literature¹⁷. This comparative analysis was carried out in order to validate a CFD code for a radial load variation range extending from 2000 to 10.000 N.

When the bearing is operated at a low-speed of 1000 rpm, under a load of 10.000 N, significant deformation is observed at the ring, as shown in Fig. 8. This graphical representation highlights the relative displacement for both the experimental and numerical approaches, revealing a proportional increase in the relative displacement of the inner face of the ring with increasing load. The results obtained using the EHD (Elastohydrodynamic Lubrication) approach show reasonable agreement with the experimental data, with an estimated deviation of only 5%.

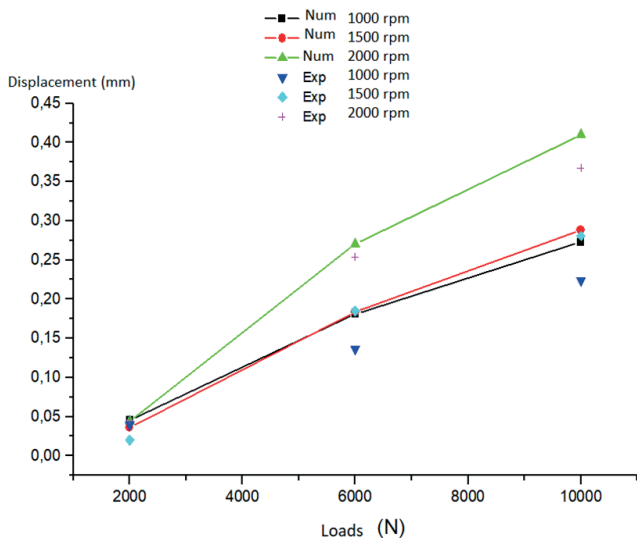


Fig. 8. Experimental and numerical displacement of the inner face of the bearing as a function of load

RESULTS AND DISCUSSION

Coated plain bearing: Rotational speeds effects. In this section, we examine the effect of shaft rotational speed on the pressure distribution in the midplane of a journal bearing. The shaft speed is varied from 6.000 to 12.000 rpm, with supply pressure $P_a = 0.08$ MPa, supply temperature $T_a = 40^\circ\text{C}$ and radial load $W = 10.000$ N.

Figure 9 shows the evolution of the hydrodynamic pressure for different rotational speeds. This curve shows that the pressure increases proportionally with

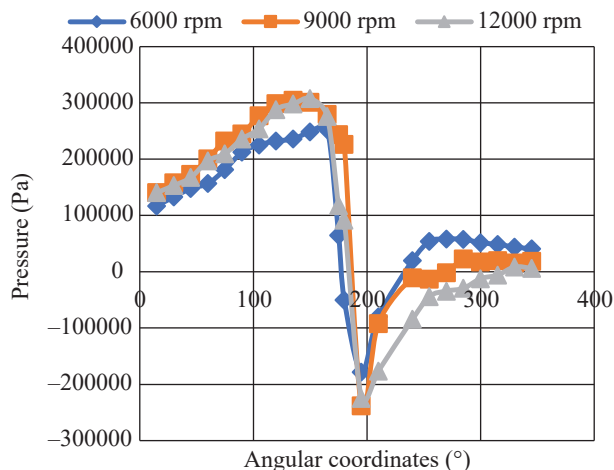


Fig. 9. Pressure distribution according to angular position for different rotational speeds

increasing shaft speed. Significant pressure values are observed for a rotation speed of 12.000 rpm, reaching a maximum at an angular position of 150°. A notable observation is that, at an angular position of 200°, the pressure is lower than the supply pressure. This is because this zone corresponds to the region where the oil film breaks. At this position, the pressure exerted on the plain bearing is not sufficient to maintain a continuous film of oil between the bearing and the shaft, resulting in a decrease in the pressure observed. The distribution of displacement, shear stress, and elastic strain due to the pressure field for a tin-coated plain bearing are shown in Fig. 10.

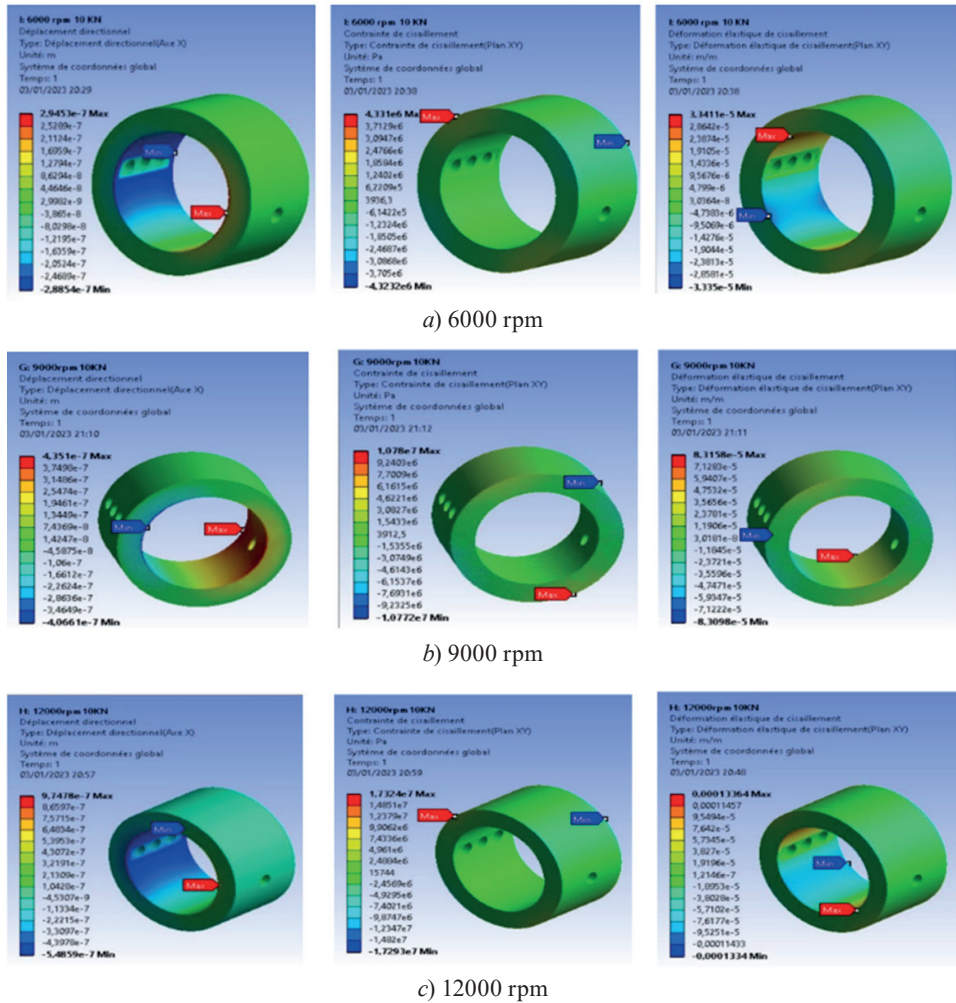


Fig. 10. Evolution of displacement, shear stress, and elastic strain for different rotational speeds. (Tin-coating plain bearing)

Figure 11 shows the displacement distribution of the inner faces of the bearing as a function of angular position for rotational speeds of 6,000, 9,000, and 12,000 rpm, for both the coated plain bearing and the composite plain bearing. The effect of increasing speed on the elastic parameters of the plain bearing can be clearly seen in this graph.

Increasing the speed of rotation leads to an increase in the displacements, shear stresses, and elastic deformations of the plain bearing. At a speed of 12,000 rpm, these parameters all become significant, indicating a greater mechanical load on the bearing.

It is interesting to note that the maximum displacement is observed at angular positions between 150° and 250°, corresponding to the lower generatrix of the bearing. This observation suggests that the most stressed area of the bearing is located in this region, where mechanical forces and stresses are at their highest due to the specific operating configuration of the bearing.

The analysis reveals a significant increase in displacement with increasing speed, varying according to the type of coating or composite. For tin-coated bearings, this increase is of the order of 71%, while for carbon-epoxy composite bear-

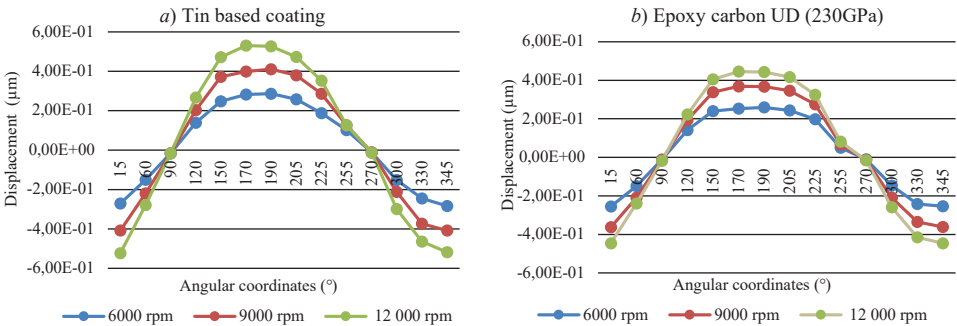


Fig. 11. Displacement according to angular position for different rotational speeds, $W = 10,000 \text{ N}$

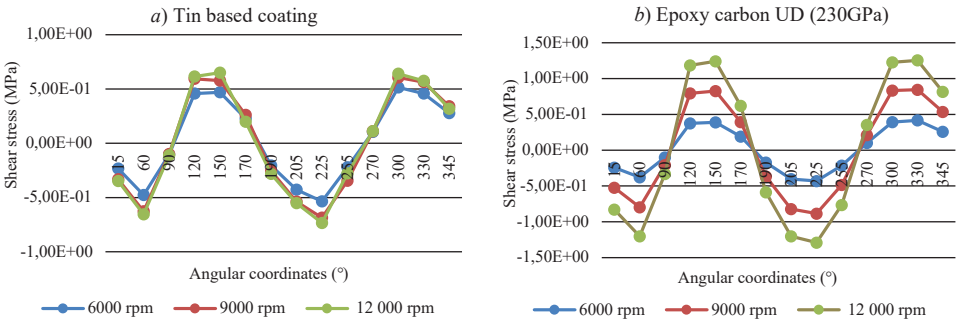


Fig. 12. Shear stress according to angular position for different rotational speeds, $W = 10,000 \text{ N}$

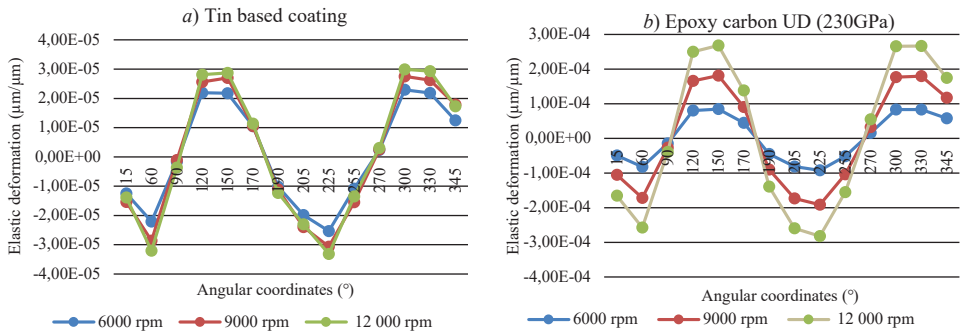


Fig. 13. Elastic deformation according to angular position for different rotational speeds, $W = 10,000 \text{ N}$

ings, it is no more than 40%. This difference highlights the distinct mechanical properties of the materials used, influencing their behaviour under load and their ability to absorb mechanical stresses.

Shear stresses and elastic strains become significant at a rotational speed of 12,000 rpm, with maximum values observed at angular positions of 150° and 330° , as shown in Figs 12 and 13. This observation highlights the critical impact of rotational speed on the mechanical stresses experienced by plain bearings, particularly in areas of maximum load.

An increase in speed also leads to an increase in shear stress, estimated at 51% for coated plain bearings, whereas this increase is much smaller, only 6%, for composite plain bearings. This difference highlights the ability of composites to reduce mechanical stresses compared with traditional metal coatings.

Also, the increase in elastic deformation is significant, reaching around 45% for coated bearings and 66% for composite bearings, as shown in Fig. 13. This difference highlights the ability of composites to absorb and dissipate mechanical stresses more effectively than metal coatings, offering greater resistance to fatigue and permanent deformation.

Different coatings (tin-lead-bronze). Figure 14 shows the displacement distribution along the median plane of a plain bearing coated with tin, lead and bronze, subjected to a load of 10,000 N and a speed of 6,000 rpm. It was observed that the displacement varied considerably depending on the type of coating applied. In particular, the tin-coated bearing shows significant displacement, reflecting the high mechanical stresses to which it is subjected. In contrast, for the bronze-coated bearing, the displacement is less pronounced, particularly in the 190° position, with an estimated difference of 64%.

The distribution of the shear stress as a function of the angular position of the plain bearing is illustrated in Fig. 15, for different coatings such as tin, lead and

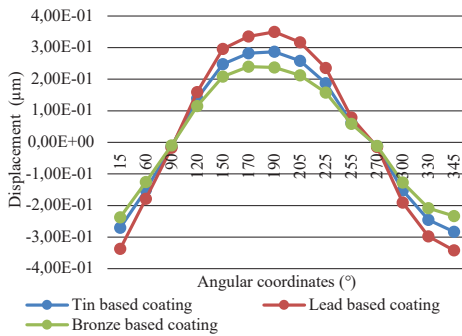


Fig. 14. Displacement according to angular position for different coatings, $N = 6,000$ rpm, $W = 10,000$ N

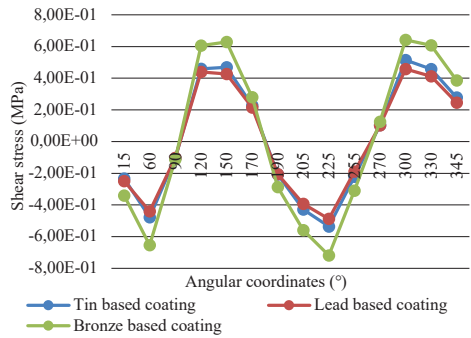


Fig. 15. Shear stress according to angular position for different coatings, $N = 6,000$ rpm, $W = 10,000$ N

bronze, at a rotation speed of $N = 6,000$ rpm and a radial load of $W = 10,000$ N. It can be seen that the shear stresses are lower for the bronze-coated bearing compared with the other coatings. This difference highlights the specific mechanical properties of bronze, which offer better resistance to shear stresses under dynamic operating conditions.

Figure 16 shows the distribution of elastic deformation along the median plane of the plain bearing for different coatings. The curves clearly indicate that the bronze-coated plain bearing deforms less than the tin- or lead-coated bearings, with a difference of around 29%. This observation highlights the importance of the choice of coating in the design of plain bearings, in particular to ensure adequate resistance to elastic deformation under load.

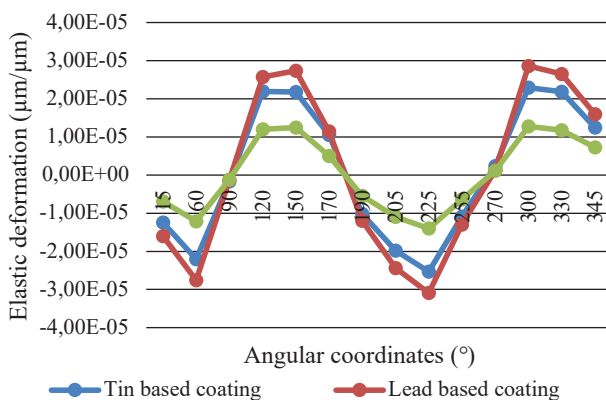


Fig. 16. Elastic deformation according to angular position for different coatings, $N = 6,000$ rpm, $W = 10,000$ N

Different composite materials. Figure 17 gives an overview of the displacement distribution along the median plane of boron/epoxy, graphite/epoxy and carbon/epoxy UD plain bearings, subjected to a load of 10,000 N and a speed of 6,000 rpm. It can be seen that the displacement varies considerably depending on the sliding bearing material.

In particular, for the UD epoxy carbon sliding bearing, the displacement is significant, reflecting the high mechanical stresses to which it is subjected. This result can be attributed to the intrinsic properties of UD epoxy carbon, such as its rigidity and strength, which can induce more pronounced deformations under load. In contrast, for the graphite/epoxy plain bearing, the displacement appears to be less significant, particularly at the 190° position. This observation suggests that the graphite/epoxy composite material offers better mechanical stability and increased resistance to deformation, even under high loads and at high speeds.

Figure 18 shows the distribution of shear stresses as a function of angular position of the plain bearing, for different composite materials such as boron/epoxy, graphite/epoxy and UD epoxy carbon, at a rotational speed of $N = 6,000$ rpm and a radial load of $W = 10,000$ N. It is observed that the shear stresses are lower for the boron/epoxy bearing compared with the other composite materials. This difference can be attributed to the specific mechanical properties of boron/epoxy, such as its strength and rigidity, which help to reduce the shear stresses in the plain bearing.

Figure 19 shows the distribution of elastic deformation along the median plane of the plain bearing for the different composite materials. It can be seen that the boron/epoxy sliding bearing has less elastic deformation than the UD epoxy carbon or graphite/epoxy bearings. This observation suggests that boron/epoxy offers better resistance to deformation under load, which may contribute to better stability and durability of the plain bearing under dynamic operating conditions.

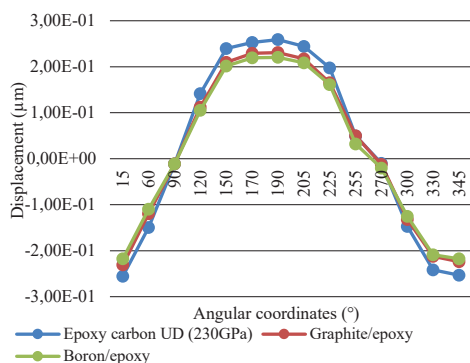


Fig. 17. Displacement according to angular position for different composite materials, $N = 6,000$ rpm, $W = 10,000$ N

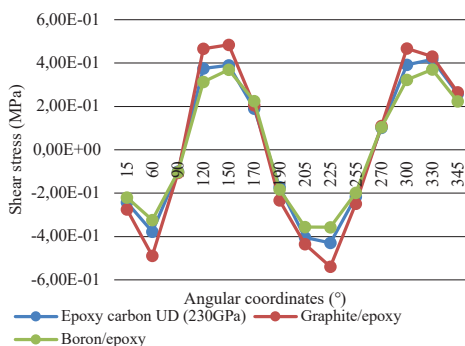


Fig. 18. Shear stress according to angular position for different composite materials, $N = 6,000$ rpm, $W = 10,000$ N

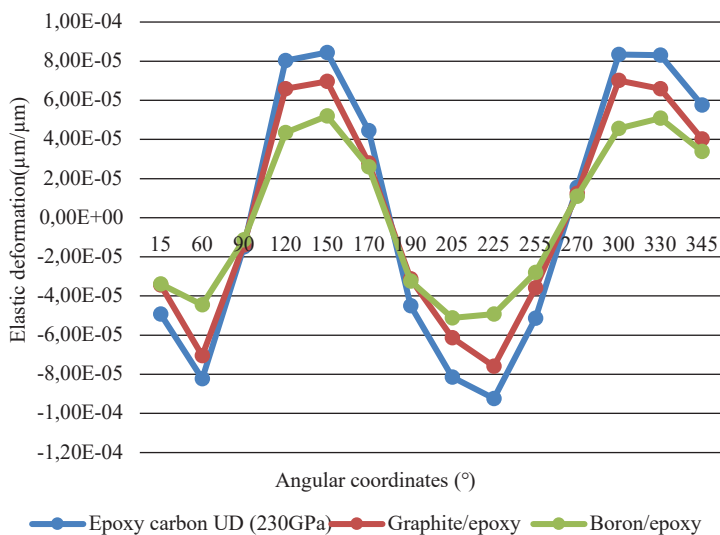


Fig. 19. Elastic deformation according to angular position for different composite materials, $N = 6,000$ rpm, $W = 10,000$ N

The importance of the choice of composite material in the design and optimization of plain bearings to ensure their performance and reliability in demanding industrial applications. By selecting the right composite material, it is possible to improve the mechanical properties of the plain bearing, particularly by reducing shear stresses and elastic deformation, which can lead to improved efficiency and longer life for the system as a whole.

Results comparison. Figure 20 shows the variation of structural parameters along the median plane of the plain bearing, such as displacement, shear stress and elastic strain. It can be seen that the displacement of the inner face of the bronze-based coated sliding bearing is slightly higher than that obtained for a composite (boron/epoxy) sliding bearing, with a difference of around 8% (see Fig. 20a). As far as shear stress is concerned, it is significantly higher for a coated bearing than for a composite bearing (see Fig. 20b). The difference is around 41%.

The elastic strain for a bronze-based coated bearing is lower than that obtained for a composite (boron/epoxy) bearing, with a difference of about 71% (see Fig. 20c).

These observations highlight the significant differences in mechanical behavior between coated plain bearings and composite bearings. Although the displacement is slightly higher for the coated bearings, the shear stresses are considerably higher, which can have important implications for the durability and reliability of the bearings under real operating conditions. Similarly, elastic deformation,

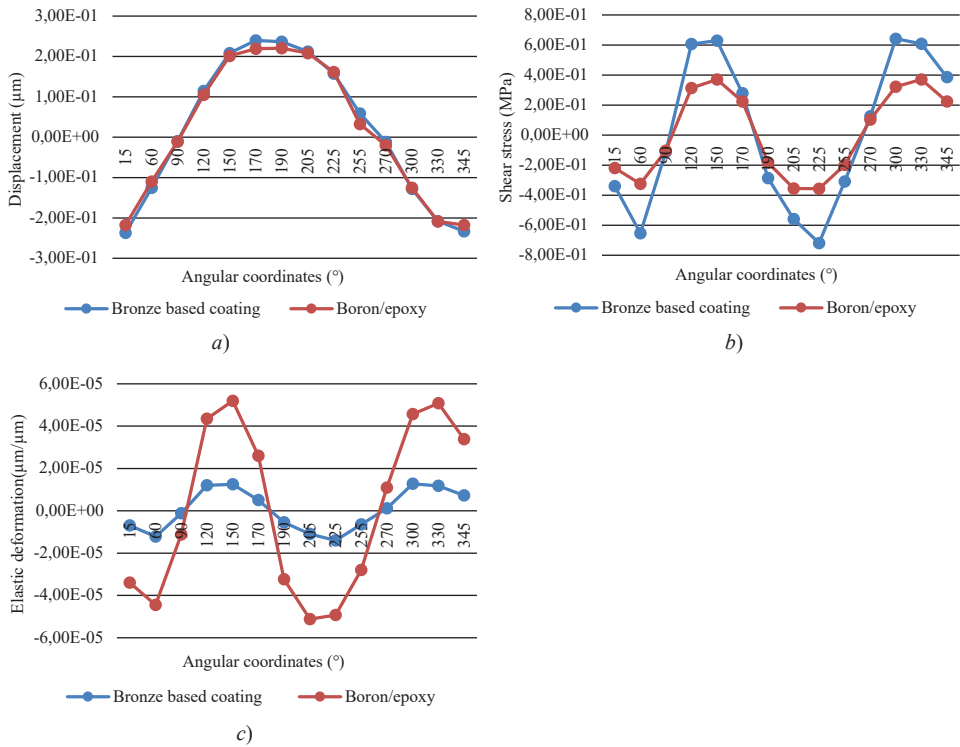


Fig. 20. Structural parameters according to the angular coordinates for coated and composite plain bearing a) Displacement; b) Shear stresses; c) Elastic deformation

although slightly lower for coated bearings, is still significant and must be taken into account when designing and optimizing bearings to ensure optimum performance in a variety of industrial applications.

CONCLUSIONS

This study provides valuable insights into the mechanical performance of hydrodynamic plain bearings as a function of the coatings and composite materials used. These results can guide design and material choices to optimise bearing performance and durability in a variety of industrial applications. This in-depth study highlights the significant impact of different coatings and composite materials on the mechanical performance of a hydrodynamic plain bearing. The main results are as follows:

- Displacement and maximum pressure: The analysis reveals that the displacement induced by the pressure field is particularly high in the angular positions of the plain bearing, notably between 120° and 250°, where the pressure is

maximum and the thickness of the oil film is minimum. This observation underlines the critical importance of these positions in the operation of the bearing.

- **Influence of hydrodynamic pressure:** The results highlight the significant importance of the hydrodynamic pressure generated by fluid movement on the displacement and shear stress of the coated plain bearing, compared with a composite plain bearing. This observation highlights the crucial impact of lubrication conditions on the mechanical performance of the bearing.

- **Effect of rotational speed:** It has been shown that an increase in rotational speed leads to a simultaneous increase in the shear stress and elastic displacement of the plain bearing. This relationship highlights the significant influence of rotational speed on the mechanical stresses experienced by the bearing.

- **Adaptability of the bronze coating:** The results indicate that the bronze-based coating adapts very well to the working conditions of the plain bearing. This observation suggests that this type of coating could be particularly useful for alleviating friction problems over a wide range of rotational speeds.

- **Elastic behaviour of the composite bearing:** In comparison, the composite sliding bearing, made of boron and epoxy, exhibits more favourable elastic behaviour with variation in rotational speed. This underlines the potential advantages of composite materials in terms of flexibility and mechanical resilience.

- **Relative deformation:** Finally, it was observed that the bronze-coated plain bearing had lower deformation than the composite (boron/epoxy) plain bearing. This difference in mechanical properties may have important implications for the durability and reliability of bearings in real applications.

These results highlight the importance of material selection in plain bearing design, in particular to ensure optimum structural stability and adequate mechanical performance under a variety of operating conditions. By selecting the appropriate material, it is possible to optimise the mechanical behaviour of the plain bearing, thereby helping to improve the reliability and durability of the mechanical systems in which they are used.

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